

Contact

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(LinkedIn)

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Top Skills

Temporal dynamics modeling

Action anticipation

OpenCV

Languages

French (Limited Working)

Spanish (Native or Bilingual)

English (Full Professional)

Catalan (Native or Bilingual)

Certifications

B2 First – Cambridge English

DELFB1

Curs de Categoria Sènior (2n Nivell Federatiu)

Honors-Awards

Best Project - AI4ALL 2023

Publications

Revealing hidden bioimaging information by isotropic depolarization filtering

Arnau Barrera Roy

PhD student in Computer Vision @UB-CVC.
Greater Barcelona Metropolitan Area

Summary

Physicist exploring the intersection of Artificial Intelligence and Sports Analytics, focused on bringing Computer Vision into the game.

Currently I am a 1st year PhD student at University of Barcelona and the Computer Vision Center, working on Knowledge Distillation of Multimodal Large Language Models.

I've done research in optics, investigating image enhancement through light polarization. I've worked as a Software Developer at the SEAT Technical Center and I'm the Co-founder of Sinfon.ia, a startup based on AI to teach you how to play your favorite songs with the piano.

In the evenings, I set my computer aside, grab my whistle and become a youth basketball coach.

Experience

Computer Vision Center

1 year 8 months

PHD Student

October 2025 - Present (11 months)

PhD student at the Human Pose and Behavior Analysis group (HuPBA) at the Computer Vision Center and University of Barcelona, under the supervision of Profs. Sergio Escalera and Albert Clapés.

During the first year, my research focuses on developing knowledge distillation methods for multimodal large language models (MLLMs). This work, in collaboration with Huawei, aims to improve multimodal alignment, generalization, and knowledge transfer capabilities in MLLMs.

Intern - Human action anticipation

January 2025 - September 2025 (9 months)

During my Master's thesis, I worked on action anticipation in basketball broadcast videos under the supervision of Prof. Albert Clapés. Key contributions of my work include:

- Creation of a large-scale dataset with 100,000+ video clips (~300 hours of footage) and more than 2,000 manually annotated rebound events
- Development of multiple deep learning approaches to anticipate offensive vs. defensive rebounds, including 3D CNNs, Vision Transformers, and RNN-based models
- ## Formulation of two anticipation settings — offline (predicting events seconds before they happen) and online (predicting imminent events without knowing their exact timing)
- Human–AI comparison study, benchmarking model performance against expert annotations
- Interpretability analysis to better understand the decision process of the models

Rebound anticipation is particularly challenging due to dense multi-player interactions, frequent occlusions, and highly irregular ball trajectories. This work demonstrates both the feasibility and the limitations of forecasting outcomes in complex multi-agent sports scenarios.

Sinfon.ia

Co-Founder

September 2023 - Present (3 years)

Federació Catalana de Basquetbol

Basketball Coach

August 2020 - Present (6 years 1 month)

Coached youth teams, both boys and girls, ranging in age from 10 to 18 years old.

Key Responsibilities:

- Plan and conduct training sessions focused on developing technical skills, tactical understanding, and physical conditioning appropriate to each age group.
- Design season-long development plans, including goals for individual and team progress.

- Foster a positive team culture by teaching core values such as respect, responsibility, and collaboration, helping young players grow both as athletes and individuals.
- Manage game-day preparation, including strategy, line-ups, and in-game adjustments.
- Evaluate player performance and provide constructive feedback to support growth and motivation.
- Communicate regularly with parents, club staff, and other coaches to ensure alignment on objectives and player development.

SEAT S.A.

Software Developer

September 2023 - August 2024 (1 year)

Responsible for analyzing client requirements, designing, developing, and maintaining customized management and planning software for SEAT's Technical Center using MATLAB and App Designer. The software supports efficient team, project, and testing management tailored to the unique needs of each laboratory while ensuring integration compatibility with other systems.

Key Responsibilities:

- Develop and customize both the front end and back end modules to manage teams, projects, and testing operations effectively.
- Collaborate closely with end-users to understand requirements and adapt the software accordingly.
- Integrate data and automate workflows to improve operational efficiency and decision-making.
- Provide ongoing support, updates, maintenance, and troubleshooting.
- Assist in user training to maximize adoption and effective use of the software.

Universitat Autònoma de Barcelona

Research Intern

December 2022 - August 2023 (9 months)

Department of Physics, Optics Group.

During my time with the Optics Group at UAB, I focused on developing computational methods to enhance the contrast of polarimetric images of biological samples and evaluate their performance. My main contributions included:

- Polarimetric analysis: Applied various polarimetric decompositions derived from Mueller matrices and identified observables with the highest contrast in tissue images.
- Design of polarimetric filters: Developed and implemented a polarimetric filter to improve contrast in biological images, facilitating the detection of specific structures and damaged areas in tissues.
- Programming and image processing: Calibrated optical systems, measured biological samples, and created custom analysis tools in Python and MATLAB.
- Machine learning for tissue classification: Trained a neural network using a polarimetric dataset to classify four types of tissues: muscle, tendon, bone, and myotendinous junction.

I also co-authored a scientific article titled “Revealing hidden bioimaging information by isotropic depolarization filtering”.

CIRCUTOR

Technical assistant

June 2018 - August 2018 (3 months)

Education

Universitat de Barcelona

MSc in Computer Vision, Artificial Intelligence · (September 2024 - September 2025)

Universitat Autònoma de Barcelona

BSc in Physics · (September 2019 - September 2023)